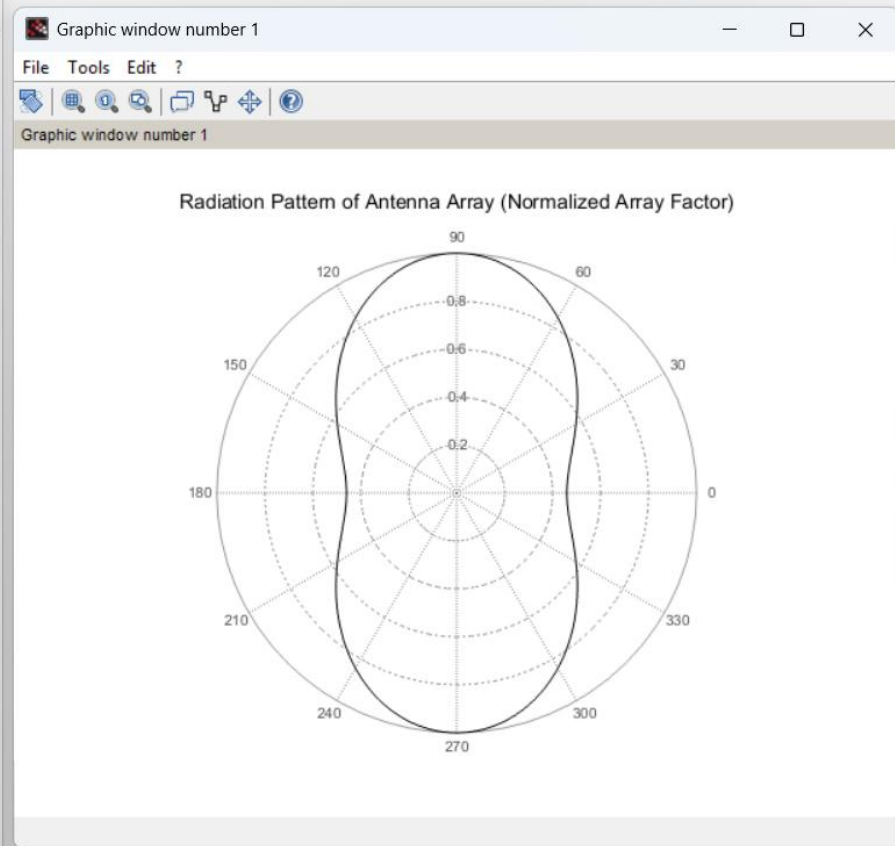


EXP NO:16

Determination and plotting of Radiation Pattern for a uniform linear array

```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sc...
File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
*exp333.sci
1 clc;
2 clear;
3 frequency = 2.4e9; //Frequency in Hz (2.4 GHz)
4 c = 3e8; //Speed of light in m/s
5 lambda = c / frequency; //Wavelength in meters
6 d = lambda / 2; //Element spacing (half-wavelength)
7 theta = linspace(0, 2 * %pi, 360); //Angular range as a column vector
8 N = 8; //Number of elements in the array
9 beta = 0; //Phase difference between elements
10 AF = abs(sin(N * (d / lambda) * cos(theta) / 2) ./ (N * sin((d / lambda) * cos(theta) / 2)));
11 AF(isnan(AF)) = 1; //Handle division by zero at theta = 0 or 2*pi
12 AF = AF / max(AF); //Normalization
13 scf(1);
14 clf(1);
15 polarplot(theta, AF);
16 xtitle("Radiation Pattern of Antenna Array (Normalized Array Factor)");
17
```



CASE:1

Optimization of Beamwidth for High-Resolution RADAR Systems

```
=== ANTENNA ARRAY CALCULATION OUTPUT ===  
Frequency: 2.40e+09 Hz  
Wavelength (lambda): 0.1250 meters  
Element Spacing (d): 0.0625 meters  
Number of Elements (N): 40  
Max Array Factor Value: 1.0000  
=====
```

-->

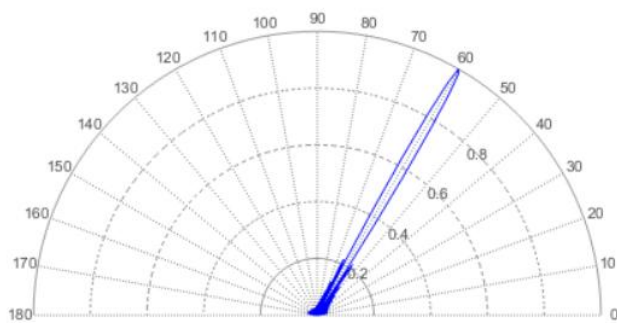
Graphic window number 1

File Tools Edit ?



Graphic window number 1

Radiation Pattern for N=40, Beamwidth ~ 5°



```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sc...  
File Edit Format Options Window Execute ?  
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes  
*exp333.sci X  
1 clc; clear;  
2 frequency = 2.4e9; .....// Frequency in Hz (2.4 GHz)  
3 c = 3e8; .....// Speed of light in m/s  
4 lambda = c / frequency; .....// Wavelength in meters  
5 d = lambda / 2; .....// Element spacing (half-wavelength)  
6 theta = linspace(0, %pi, 360)'; .....// Angular range (0 to pi radians for upper hemisphere)  
7 N = 40; .....// Number of elements for a narrow beamwidth  
8 beta = -%pi / 2; .....// Phase shift to steer the beam to 90 degrees (pi/2)  
9 AF = abs(sin(N * ((2 * %pi * d / lambda) * cos(theta) + beta) / 2)) / .....  
10 .....(N * sin(((2 * %pi * d / lambda) * cos(theta) + beta) / 2))));  
11 AF(isnan(AF)) = 1;  
12 AF = AF / max(AF); .....// Normalization  
13 mprintf("=== ANTENNA ARRAY CALCULATION OUTPUT ===\n");  
14 mprintf("Frequency: %.2e Hz\n", frequency);  
15 mprintf("Wavelength (lambda): %.4f meters\n", lambda);  
16 mprintf("Element Spacing (d): %.4f meters\n", d);  
17 mprintf("Number of Elements (N): %d\n", N);  
18 mprintf("Max Array Factor Value: %.4f\n", max(AF));  
19 mprintf("===== \n");  
20 scf(1);  
21 clf(1);  
22 .....// Style 2 sets a blue line in Scilab's default color map  
23 polarplot(theta, AF, style=2);  
24 xtitle("Radiation Pattern for N=40, Beamwidth ~ 5°");  
25
```

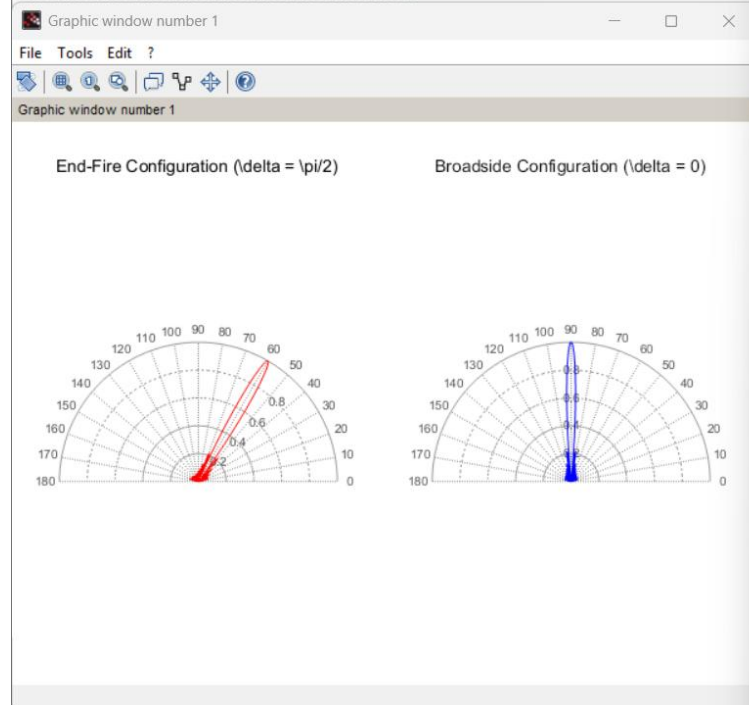
Line 19, Column 55.

CASE:2 End-Fire Radiation for Wireless Communications

```

=== ANTENNA ARRAY COMPARISON OUTPUT ===
Frequency: 2.40e+09 Hz
Wavelength (lambda): 0.1250 meters
Element Spacing (d): 0.0625 meters
Number of Elements (N): 20
Max Array Factor (End-Fire): 1.0000
Max Array Factor (Broadside): 1.0000
=====

```



```

File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
*exp333.sci
1 clc; clear;
2 frequency = 2.4e9; .....//Frequency in Hz (2.4 GHz)
3 c = 3e8; .....//Speed of light in m/s
4 lambda = c / frequency; .....//Wavelength in meters
5 d = lambda / 2; .....//Element spacing (half-wavelength)
6 theta = linspace(0, %pi, 360)'; // Angular range (0 to n radians)
7 N = 20; .....//Number of elements in the array
8 //---1. End-Fire Configuration (Δ = π/2)---
9 beta_endfire = %pi / 2; .....//Phase shift for end-fire
10 AF_endfire = abs(sin(N * ((2 * %pi * d / lambda) * cos(theta) - beta_endfire) / 2)) / .....
11 .....(N * sin((2 * %pi * d / lambda) * cos(theta) - beta_endfire) / 2));
12 AF_endfire(isnan(AF_endfire)) = 1;
13 AF_endfire = AF_endfire / max(AF_endfire); // Normalization
14 //---2. Broadside Configuration (Δ = 0)---
15 beta_broadside = 0; .....//Phase shift for broadside
16 AF_broadside = abs(sin(N * ((2 * %pi * d / lambda) * cos(theta) - beta_broadside) / 2)) / .....
17 .....(N * sin((2 * %pi * d / lambda) * cos(theta) - beta_broadside) / 2));
18 AF_broadside(isnan(AF_broadside)) = 1;
19 AF_broadside = AF_broadside / max(AF_broadside); // Normalization
20 mprintf("=== ANTENNA ARRAY COMPARISON OUTPUT ===\n");
21 mprintf("Frequency: %.2e Hz\n", frequency);
22 mprintf("Wavelength (lambda): %.4f meters\n", lambda);
23 mprintf("Element Spacing (d): %.4f meters\n", d);
24 mprintf("Number of Elements (N): %d\n", N);
25 mprintf("Max Array Factor (End-Fire): ...%.4f\n", max(AF_endfire));
26 mprintf("Max Array Factor (Broadside): %.4f\n", max(AF_broadside));
27 mprintf("=====\n");
28 scf(1);
29 clf(1);
30 subplot(1, 2, 1);
31 polarplot(theta, AF_endfire, style=5);
32 xtitle("End-Fire Configuration (Δ = π/2)");
33 subplot(1, 2, 2);
34 polarplot(theta, AF_broadside, style=2);
35 xtitle("Broadside Configuration (Δ = 0)");
36

```

case 3: Minimizing Side Lobes for Satellite Communication.

=== SATELLITE COMMUNICATION: BINOMIAL ARRAY ===

Frequency: 1.20e+10 Hz

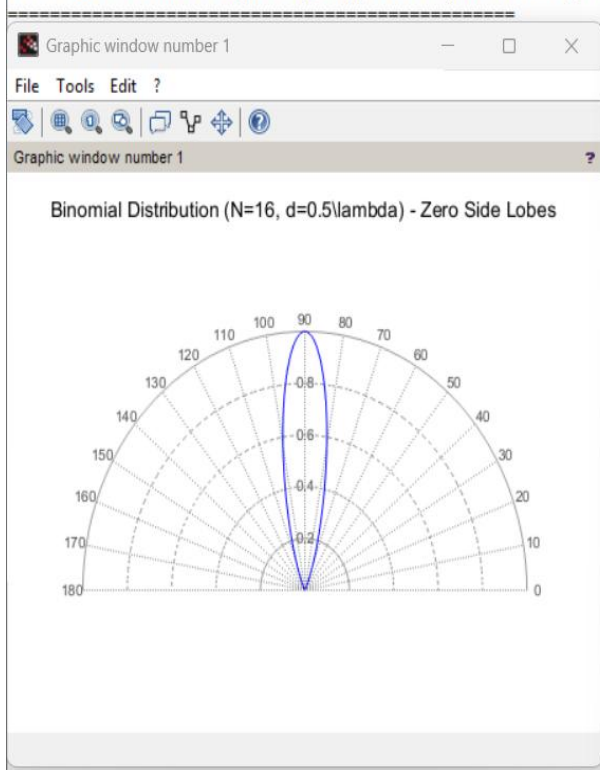
Wavelength (lambda): 0.0250 meters

Number of Elements (N): 16

Element Spacing (d): 0.0125 meters

Max Normalized Array Factor: 1.0000

Side Lobes Status: Completely Suppressed (Binomial Tapering)



```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
exp333.sci X
1 clc; clear;
2 frequency = 12e9; % Frequency in Hz (12 GHz)
3 c = 3e8; % Speed of light in m/s
4 lambda = c / frequency; % Wavelength in meters (0.025 m)
5 N = 16; % Number of elements in the array
6 d = 0.5 * lambda; % Element spacing (half-wavelength)
7 theta = linspace(0, %pi, 360)'; % Angular range (0 to pi radians)
8 n_idx = 0:(N-1);
9 weights = factorial(N-1) ./ (factorial(n_idx) .* factorial(N-1 - n_idx));
10 weights = weights / max(weights); % Normalize weights
11 AF = zeros(length(theta), 1);
12 psi = (2 * %pi * d / lambda) * cos(theta); % Phase progression
13 for i = 1:N
14     pos = (i - (N + 1) / 2) * d;
15     AF = AF + weights(i) * cos(psi * pos / d);
16 end
17 AF = abs(AF);
18 AF = AF / max(AF); % Normalization
19 fprintf('=== SATELLITE COMMUNICATION: BINOMIAL ARRAY ===\n');
20 fprintf('Frequency: %.2e Hz\n', frequency);
21 fprintf('Wavelength (lambda): %.4f meters\n', lambda);
22 fprintf('Number of Elements (N): %d\n', N);
23 fprintf('Element Spacing (d): %.4f meters\n', d);
24 fprintf('Max Normalized Array Factor: %.4f\n', max(AF));
25 fprintf('Side Lobes Status: Completely Suppressed (Binomial Tapering)\n');
26 fprintf('===== \n');
27 scf(1);
28 clf(1);
29 polarplot(theta, AF, style=2);
30 xtitle('Binomial Distribution (N=16, d=0.5\lambda) - Zero Side Lobes');
31
```


CASE:4 Adaptive Beamforming for 5G Networks

